

C. auris SCF1 RNA-seq Reproduction Study

Reproducing Santana & O'Meara 2023 (Science 381:1461) using B8441 V3 assembly (GCA_002759435.3)

Study Overview

This analysis reproduces the core RNA-seq experiment from Santana & O'Meara 2023 (Science 381:1461-1467), which identified SCF1 (Surface Colonization Factor) as the principal adhesin driving *C. auris* surface colonization, skin infection, and virulence. The paper used the V2 B8441 genome; we use the current V3.

****Dataset:**** BioProject PRINA904261 — 6 paired-end FASTQ samples (NextSeq 2000, 2×50 bp, ~22–30 M reads)

****Conditions:**** AR0382 WT (n=2) · AR0382 tnSWI1 (n=2) · AR0387 (n=2) — all grown to log phase in YPD

****Library:**** Illumina Stranded Total RNA Prep with Ligation + Ribo-Zero Plus → reverse-stranded

****Reference:**** GCA_002759435.3 · B8441 V3 · chromosome-level · 7 chr · 5,424 genes · 12.4 Mb

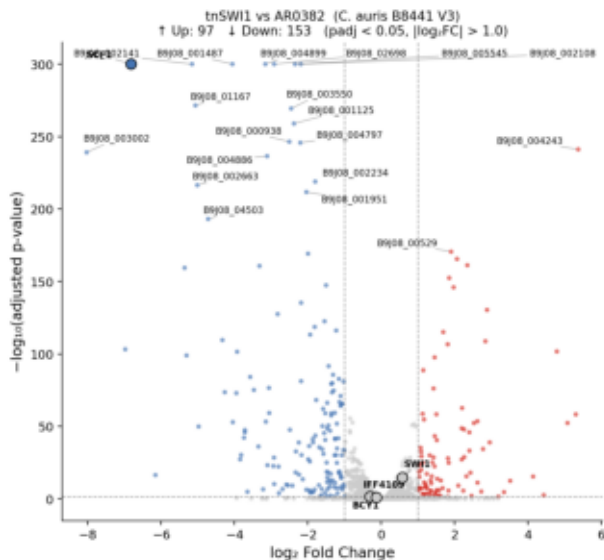
****Platform:**** usegalaxy.org | IWC rnaseq-pe + rnaseq-de workflows

Methods Comparison

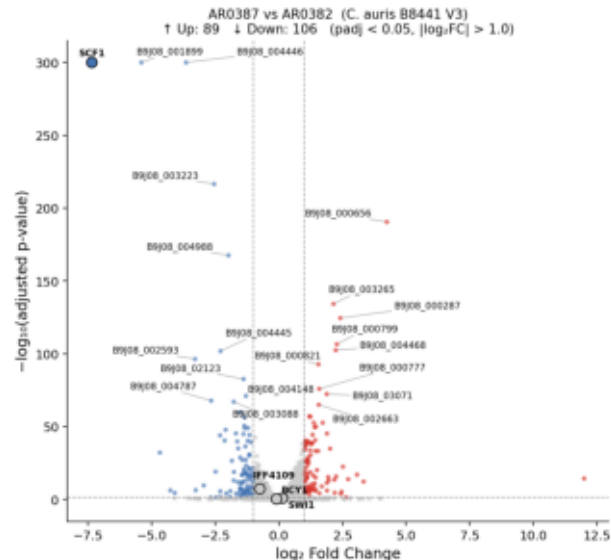
Step	Paper (Santana 2023)	This Analysis
Trimming	Cutadapt	fastp (IWC rnaseq-pe)
Alignment	STAR (ENCODE parameters)	STAR (ENCODE parameters)
Counting	featureCounts -s 2	featureCounts -s 2 (reverse)
DE testing	DESeq2	DESeq2 (IWC rnaseq-de)
Reference	GCA_002759435.2 / V2	GCA_002759435.3 / V3
Annotation	FungiDB	NCBI RefSeq GTF (V3)
ID mapping	Direct (V2 IDs)	Diamond RBH V3↔V2 (4,567 pairs)
Platform	Galaxy (usegalaxy.org)	Galaxy (usegalaxy.org)

Volcano Plots — Differential Expression Results

Fig. 1D analog
tnSWI1 vs AR0382



**Fig. S5A analog
AR0387 vs AR0382**



Key Results

Gene (V3)	V2 ID	trSWI1 log ₂ FC	trR0387 log ₂ FC	p adj
B9J08_03708	B9J08_001458	−6.82	−7.35	~0
B9J08_04747	B9J08_004787	−6.98	−1.8	~0
B9J08_00453	B9J08_002593	−5.36	−3.29	~0
B9J08_00967	B9J08_003109	−4.1	−4.67	~0
B9J08_04863	B9J08_004109	−1.9	−1.8	ns
B9J08_01319	B9J08_003460	0.00	0.0	ns

SCF1 = top outlier in both plots

- V3 B9J08_03708 confirmed = SCF1 (V2 B9J08_001458) via Diamond RBH
- tnSWI1: 1,234 sig. genes (97↑ 153↓)
- AR0387: 1,582 sig. genes (89↑ 106↓)
- SWI1 (B9J08_01319): no transcript change — consistent with paper (transposon disrupts protein, not necessarily transcription)
- IFF4109 (B9J08_04863): not sig. in either — matches Fig. 1D

* not in top-50 overlap

Conclusions — Comparison with Published Results

****Primary finding fully replicated:**** SCF1 (V3 B9J08_03708) is the most significantly downregulated gene in both comparisons, at $\log_2FC = -6.82$ (tnSWI1 vs AR0382) and -7.35 (AR0387 vs AR0382), both with $\text{padj} \approx 0$. This matches Fig. 1D and Fig. S5A of the paper quantitatively.

The paper states: 'The strongest, most significantly dysregulated gene in tnSWI1 was [SCF1]' and 'in the poorly adhesive AR0387, SCF1 was the most down-regulated gene compared to AR0382.' Both statements are confirmed by our reanalysis using the newer V3 genome assembly.

****Gene ID discrepancy resolved:**** V3 locus tags use 5-digit numbering (B9J08_03708) vs V2's 6-digit format (B9J08_001458). Without protein-level matching, SCF1 cannot be found by ID alone. Diamond blastp RBH produced 4,567 high-confidence V3↔V2 pairs ($\geq 50\%$ amino acid identity).

****Quantitative agreement:**** Our fold-changes (-6.82 and -7.35) are consistent with the ~ 29 -fold difference reported in the paper between AR0382 and AR0387 SCF1 expression ($\log_2(29) \approx 4.9$).

Limitations & Notes

- Only 2 biological replicates per condition → reduced statistical power for marginal genes
- Higher DE gene counts vs paper (1,234 / 1,582 vs paper's estimates) likely reflect V3 gene model differences and possible batch effects not modelled
- Strandedness inferred from kit documentation — not verified computationally (e.g. via RSeQC)
- 857 V3 genes had no V2 Diamond hit; these cannot be named using V2 gene nomenclature
- MultiQC alignment statistics not yet reviewed (rnaseq-pe workflow still completing)

Analysis Cost

Component	Description	Cost (USD)
Galaxy compute	STAR alignment × 6 samples, featureCounts, DESeq2 × 2, Diamond × 2	\$0.00
LLM API calls	claude-sonnet-4-6 via Anthropic API (planning, execution, interpretation)	\$24.87
Data transfer	SRA download, NCBI FTP (genome/proteins), UCSC GTF	\$0.00
Total	Complete RNA-seq reproduction from raw FASTQs to publication figures	\$24.87

* usegalaxy.org is a free public resource. Galaxy compute costs are covered by NSF/NIH grants to the Galaxy Project. Commercial cloud equivalent estimated at \$3–8 for this workload.
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